

CULVERTCRAWL · DESIGN DECISION RECORD

CulvertCrawl TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	CVC-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/culvertcrawl		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Items 9 to 11 decided by Amish on 2026-10-02 (CVC-DEC-001)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 8); items 9 to 11 decided by Amish on 2026-10-02 (CVC-DEC-001)

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish", and the precis (CVC-PRC-001 v0.2) gave a recommendation for most of them. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."

This record lists what that instruction decides and what it leaves open because there was no recommendation to accept. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (TRL 2 section) and in CVC-PRC-001 v0.2, "Key design choices". They are not repeated here.

■ Decision

Table 1. Decided items.

#	Item	Decision	Where it now lives
1	Budget	Decided by Amish, 2026-09-25: go with recommendation. Raise <code>budget_usd</code> from \$800 to \$900, keeping the 60 m wired tether and slip ring	<code>project.yaml</code> , CVC-REQ-001 R12
2	Pipe range	Decided by Amish, 2026-09-25: go with recommendation. Keep 300 to 900 mm and check the 900 mm case early at TRL 3 (done in CVC-CAL-001: R5 not met at 900 mm)	CVC-REQ-001 R1, R5
3	Surface battery	Decided by Amish, 2026-09-25: go with recommendation. 12.8 V 20 Ah LiFePO4 with a 48 V boost for the prototype; a SwapCell-compatible input stays a later option	CVC-PRC-001 v0.3
4	Drive	Decided by Amish, 2026-09-25: go with recommendation. Tracks rather than wheels	CVC-PRC-001 v0.3
5	Power architecture	Decided by Amish, 2026-09-25: go with recommendation. Surface power over the tether at 48 V DC; no battery in the crawler	CVC-PRC-001 v0.3, CVC-REQ-001 R11
6	Profiling sensor	Decided by Amish, 2026-09-25: go with recommendation. Class 2, 520 nm laser ring ahead of a fixed fisheye camera, rather than structured light, stereo or a pan-tilt head	CVC-PRC-001 v0.3
7	Processing	Decided by Amish, 2026-09-25: go with recommendation. Ring extraction and reports on the operator's laptop; the crawler streams video and sensor data	CVC-PRC-001 v0.3
8	Pitch and problem wording	Decided by Amish, 2026-09-25: go with recommendation. Unchanged; the TRL 2 review recommended no rewording	<code>project.yaml</code> , <code>README.md</code>

Notes on the portfolio-wide approvals from the same instruction:

- **SwapCell.** CulvertCrawl does not use a SwapCell pack. If the later SwapCell-compatible input is taken up, it should cite SwapCell interface v0.3, whose item W (wake on a coded INTERLOCK loop, no CAN host needed) removes the CAN heartbeat concern raised at TRL 2. A shared SwapCell pack would be priced once in the SwapCell repo and excluded from this kit budget.
- **Co-design partners** are picked per area later; the first partner stays open (item 10).

Items that remain open

These had no recommendation to accept and stayed open. Recommendations were written for them in the design decisions register (CVC-DEC-001), and Amish approved them on 2026-10-02 ("i approve your recommendations for all 555 open decisions."):

1. **Output format first:** a PACP-style observation log, FHWA condition ratings or a deflection report. No preference was stated. Decided 2026-10-02: the deflection report first, observations in NASSCO PACP vocabulary second, FHWA condition ratings later.
2. **First partner user group** for field trials (county road department, watershed group or university transportation program). Not named. Decided 2026-10-02: a county road department, with a university transportation program as backup (first candidate to approach for the backup: the Texas A&M Transportation Institute).

3. **Removable laser boom** for tight or bent pipes. Marked "proposed" in CVC-PRC-001 v0.2 without a recommendation. Decided 2026-10-02: two bolts at the fin bracket for the prototype; a tool-free clamp to be decided after the first pipe trials (CVC-DDR-003, A2).

The new proposals from the TRL 3 calculations (R2 reach, R5 at 900 mm and R12 cost) were decided by Amish on 2026-09-25 by accepting the recommendations; see CVC-DDR-002 (docs/decisions/0002-recommendations-accepted.md). Items 9 to 11 had no recommendation then; they were decided on 2026-10-02 as above.

■ Consequences

- `project.yaml` `budget_usd` is 900. The priced BOM is \$901, so R12 is recorded as not met by \$1 (CVC-CAL-001).
- R5 keeps its 1 % target over 300 to 900 mm and is recorded as not met at 900 mm.
- CVC-PRB-001, CVC-PRC-001 and CVC-REQ-001 move to version 0.3 with these decisions recorded.
- No TRL 4 work is started.